Sensors for a Formula Student Driverless Racecar: Working Principles, Applications, Cost and Sourcing in India

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***Abstract*—A driverless Formula Student (FS) racecar must perceive a track that is marked only by coloured traffic cones, estimate its own motion at racing speed, and do both with sensors that a student team can afford. This report compiles the sensors commonly used for this purpose: inertial measurement units (IMUs), global navigation satellite system (GNSS) receivers, wheel speed sensors, monocular and stereo cameras, and light detection and ranging (LiDAR). For each sensor the physical principle of operation, the governing equations, the main error sources, and the specific role it plays in the perception and state-estimation pipeline of an FS Driverless car are explained. Representative products are compared from hobby grade to competition grade, together with indicative prices and where they can be bought, preferring Indian online stores and distributors. Simple calculations show why a 16-channel LiDAR cannot see cones beyond about 10 m, why stereo depth error grows with the square of range, and why standard GNSS is too coarse for cone-level localisation. The report ends with a recommended two-tier sensor suite for a team building its first driverless system.**

***Index Terms*—Formula Student Driverless, IMU, GNSS, RTK, wheel speed sensor, stereo camera, LiDAR, sensor fusion, autonomous racing.**

I. Introduction

In the driverless class of Formula Student competitions the car has to complete the acceleration, skidpad, autocross and trackdrive events with no human on board. The track is not known in advance; it is defined only by small blue cones on the left boundary, yellow cones on the right, and orange cones at the start, finish and timing lines. The software stack therefore has to answer two questions many times per second: *where are the cones?* and *where am I, and how fast am I moving?* The first is answered by perception, the second by state estimation, and the two are combined in simultaneous localisation and mapping (SLAM) before planning and control act on the result (Fig. 1).

Sensors on the car fall into two groups. **Exteroceptive** sensors observe the environment: cameras and LiDAR. **Proprioceptive** sensors measure the vehicle itself: the IMU, wheel speed sensors, steering angle and motor encoders. GNSS sits between the two, measuring absolute position from external satellite signals. No single sensor is enough; published systems such as AMZ Driverless (ETH Zürich) [1] and IIT Bombay Racing [2] fuse several of them. IIT Bombay's car, for example, combines two ZED 2i stereo cameras, a Velodyne VLP-16 LiDAR, an SBG Ellipse-N GNSS/INS and wheel encoders on an NVIDIA Jetson AGX Orin [2].

Sections II–VI cover the five core sensor types in turn, each in three parts: principle of operation, use case on our car, and available options with cost. Section VII briefly lists supporting sensors, Section VIII discusses fusion and a recommended suite, and Section IX summarises sourcing in India. Prices were checked online on 24 October 2026 and are indicative; foreign prices are converted at ₹95 per US$ and ₹127 per £ [17], [18], before shipping, customs duty and GST.

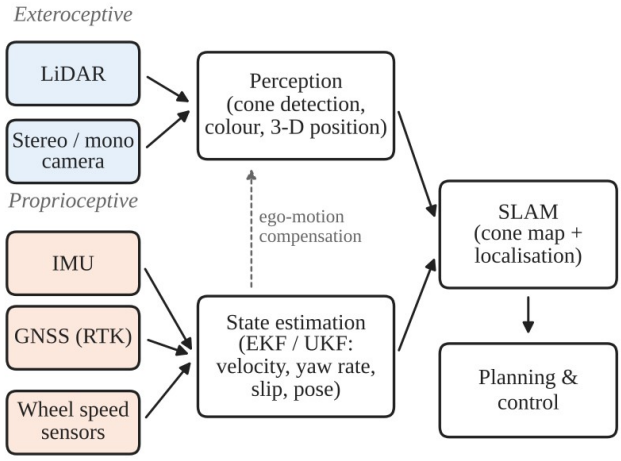


Fig. 1. Typical sensor suite and data flow in an FS Driverless car. Exteroceptive sensors feed cone perception; proprioceptive sensors feed state estimation, which also compensates LiDAR and camera data for the car's own motion.

II. Inertial Measurement Unit (IMU)

A. Principle of Operation

An IMU combines a three-axis accelerometer and a three-axis gyroscope, and often a three-axis magnetometer. In the micro-electro-mechanical systems (MEMS) devices used on student cars, the accelerometer is a tiny proof mass suspended on silicon springs; acceleration displaces the mass and the displacement is read as a change in capacitance between interleaved comb fingers. The gyroscope is a mass that is driven

to vibrate along one axis; when the sensor rotates with angular rate $\boldsymbol{\omega}$, the Coriolis force

$$\mathbf{F}_c = -2m\,(\boldsymbol{\omega} \times \mathbf{v}) \tag{1}$$

pushes the vibrating mass along a perpendicular axis, and that secondary motion, again read capacitively, is proportional to the rotation rate. Note that an accelerometer measures *specific force*, i.e. acceleration minus gravity, which is why a stationary IMU reads 9.81 m/s² upward.

Integrating gyroscope output gives orientation, and double-integrating accelerometer output (after removing gravity) gives position. The problem is that every MEMS sensor has a bias b, noise (angle/velocity random walk), scale-factor error and temperature drift. A constant accelerometer bias produces a position error

$$e(t) = \tfrac{1}{2}\, b\, t^2 \tag{2}$$

so a bias of only 0.01 m/s² (about 1 mg) grows into a 4.5 m error after 30 s. An IMU on its own is therefore excellent over short time scales but drifts over long ones, and must be corrected by other sensors. A unit that runs this correction internally—fusing the IMU with GNSS and sometimes wheel odometry in a Kalman filter—is sold as an attitude and heading reference system (AHRS) or an inertial navigation system (INS) [19].

B. Use Case on an FS Driverless Car

The IMU is the backbone of state estimation. Its yaw rate and lateral/longitudinal acceleration are the prediction (motion-model) input of the extended Kalman filter (EKF) used for velocity estimation and EKF-SLAM, as in [2]. It is the only sensor that reacts immediately to a slide, so it is also needed to estimate the vehicle side-slip angle and to drive traction control and torque vectoring. A further, less obvious use is *motion compensation*: a spinning LiDAR takes about 100 ms to complete a 10 Hz scan, during which the car can travel more than 2 m at autocross speeds, and IMU data is used to de-skew the point cloud. Mounting near the centre of gravity, rigidly and with vibration isolation from the motor, reduces lever-arm and vibration effects.

C. Options and Cost

Consumer MEMS breakouts such as the MPU-6050 (6-axis, ₹199 [6]) and the BNO085 (9-axis with on-chip sensor fusion, ₹2,279 [7]) are ideal for learning, bench tests and small-scale prototypes, but their bias instability and lack of calibration make them unsuitable as the primary sensor on the full car. Competition cars use industrial units with factory calibration and built-in GNSS fusion, for example the Xsens MTi-680G RTK GNSS/INS (≈£2,677 excluding VAT, ≈₹3.4 lakh [8]), the SBG Ellipse-N used by IIT Bombay [2], or the VectorNav VN-300 dual-antenna INS. These are sold mainly through quotation from the manufacturer or its distributors.

III. Global Navigation Satellite System (GNSS)

A. Principle of Operation

Each GNSS satellite broadcasts its orbit and a time stamp from an atomic clock. A receiver measures the signal travel time to each satellite and forms a *pseudorange* ρ_i, which contains the true distance plus the unknown offset of the receiver's cheap clock:

$$\rho_i = \|\mathbf{r}_i - \mathbf{r}\| + c\,\delta t + \varepsilon_i \tag{3}$$

With four unknowns (three coordinates of $\mathbf{r}$ and the clock bias δt), at least four satellites are needed [19]. Modern receivers track several constellations—GPS (USA), GLONASS (Russia), Galileo (EU), BeiDou (China) and India's NavIC—which improves availability. Ionospheric and tropospheric delay, satellite clock/orbit errors and multipath limit a standard single-frequency fix to roughly 1.5–3 m. Velocity, in contrast, is obtained from the Doppler shift and is accurate to a few cm/s.

Real-time kinematic (RTK) positioning removes most of the error by also measuring the phase of the carrier wave (wavelength ≈19 cm for GPS L1) and differencing it against a nearby base station at a known location, whose corrections are sent over a radio link or the internet (NTRIP). Once the integer number of carrier cycles is resolved, a dual-band receiver such as the u-blox ZED-F9P reaches centimetre-level horizontal accuracy [19]. With two antennas a fixed distance apart, the receiver can also measure true heading, independent of the magnetometer.

B. Use Case on an FS Driverless Car

FS cones are only a few metres apart and the track is narrow, so a 2 m error is useless for localisation within the track. Standard GNSS is still useful for its accurate velocity and as a slow correction for IMU drift inside an INS. RTK GNSS is a valuable *development* tool: it gives centimetre-level ground truth to evaluate SLAM and controllers during testing, and a GNSS-aided INS gives a clean velocity and heading estimate. However, GNSS reception can be degraded near buildings or stands, and the car must still drive on a track it has never seen, so the cone map produced by SLAM—not GNSS—remains the primary localisation reference [1], [2].

C. Options and Cost

A u-blox NEO-M8N module with a ceramic patch antenna is widely available from Indian hobby stores (Robu, Probots, Zbotic) for a few thousand rupees and is enough to learn the NMEA/UBX protocols and ROS 2 drivers. For RTK, the SparkFun GPS-RTK2 (ZED-F9P) board costs ₹29,482 on Robu.in [5]; a second unit (or an NTRIP service) is needed as the base station, plus multi-band antennas. Integrated RTK GNSS/INS units (Section II-C) cost several lakh rupees.

IV. Wheel Speed Sensors

A. Principle of Operation

A toothed ferromagnetic *tone ring* (or a ring of magnets) rotates with the wheel hub, and a stationary sensor produces one pulse per tooth. If the ring has N teeth, the pulse frequency is f and the effective rolling radius is r_{eff}, the wheel's linear speed is

$$v = 2\pi\, r_{\text{eff}}\, f / N \tag{4}$$

Two sensing technologies are common. A **passive inductive** (variable-reluctance) sensor is a coil around a permanent magnet; each passing tooth changes the magnetic flux and induces a voltage. It needs no power but its amplitude is proportional to speed, so it fails at very low speeds. An **active Hall-effect** (or magneto-resistive) sensor is powered and outputs a clean square wave whose amplitude does not depend on speed, so it works down to standstill. Dual-element Hall sensors give two outputs 90° out of phase (quadrature), which also reveal direction of rotation. Industrial inductive *proximity switches* are a cheap alternative that simply switch on when a metal tooth is within a few millimetres.

Resolution is set by N. With $N = 20$ and $r_{\text{eff}} = 0.23$ m, the signal is about 14 Hz at 1 m/s and over 400 Hz at 30 m/s, so speed is best measured by timing the period between pulses at low speed and counting pulses in a fixed window at high speed. On an electric car the motor's resolver or encoder, read through the inverter over CAN, provides a high-resolution speed of each driven wheel at no extra cost.

B. Use Case on an FS Driverless Car

Wheel speeds give wheel odometry, which together with the IMU forms the motion model of the EKF [2]. Comparing the wheel's circumferential speed ωr with the vehicle's true speed v gives the longitudinal slip ratio

$$s = (\omega\, r - v) / \max(\omega\, r, v) \tag{5}$$

which traction control and launch control need. Because a spinning or locked wheel no longer measures the car's speed, the estimator must detect slip and down-weight that wheel. Teams that need an independent ground-truth speed use an optical *ground speed sensor* such as the Kistler Correvit SF-Motion, which measures speed over ground optically at 500 Hz with an internal Kalman filter and was used by AMZ Racing for their 0–100 km/h record [11].

C. Options and Cost

An LJ12A3-4-Z/BX M12 inductive proximity sensor costs about ₹239 [9] and, aimed at bolt heads or a custom tone ring, is a common first wheel-speed sensor. A purpose-built Honeywell SNDH-T4L-G01 Hall-effect speed-and-direction sensor (0–15 kHz, quadrature output, −40 °C to 125 °C) is listed on IndiaMART at ₹4,500 plus tax [10]. Automotive ABS sensors from Indian two-wheeler and car spares markets are a rugged, low-cost middle option. An optical ground speed sensor such as the Kistler Correvit is a high-end, quote-based purchase.

V. Cameras: Monocular and Stereo

A. Monocular Camera — Principle

A camera focuses light through a lens onto a CMOS image sensor. Under the pinhole model, a 3-D point (X, Y, Z) in the camera frame projects to pixel (u, v) as

$$u = f_x\, X / Z + c_x, \quad v = f_y\, Y / Z + c_y \tag{6}$$

where f_x, f_y are focal lengths in pixels and (c_x, c_y) is the principal point; these *intrinsics*, together with lens-distortion coefficients, are found by checkerboard calibration [20]. A colour camera places a Bayer filter over the pixels to recover red, green and blue. For a moving car two properties matter. A **global-shutter** sensor exposes all pixels at the same instant, while a cheaper **rolling-shutter** sensor reads out row by row, so fast motion skews the image. Short exposure and good dynamic range are needed to avoid motion blur and to handle sun and shadow on the same frame.

The projection loses depth, so one image cannot directly give distance. For an object of known height H, however, its image height h gives $Z \approx f H / h$. A small FS cone is 0.325 m tall; with $f = 700$ px, a cone at 10 m is only about 23 px tall, so a 1-pixel error in the bounding box already gives about 4 % depth error. IIT Bombay Racing fitted a power-law to bounding-box height and obtained 4.5 % average error for upright cones [2].

B. Stereo Camera — Principle

A stereo camera has two synchronised, calibrated cameras separated by a baseline b. After rectification, a point appears in both images on the same row, shifted horizontally by the *disparity* $d = u_L - u_R$. Triangulation gives

$$Z = f\, b / d, \quad \delta Z \approx Z^2\, \delta d / (f\, b) \tag{7}$$

The second expression is the key limitation: depth error grows with the *square* of range. For a ZED 2i ($b = 0.12$ m) with $f = 700$ px and a quarter-pixel matching error, δZ is about 0.3 m at 10 m but about 1.2 m at 20 m (Fig. 2). A wider baseline or higher resolution improves this at the cost of a larger minimum range and more computation. Matching also fails on texture-less surfaces, and active stereo cameras such as the Intel RealSense D435i add an infrared pattern projector that helps indoors but is washed out by sunlight.

C. Use Case on an FS Driverless Car

Cameras are the only sensor that sees *colour*, which is essential because cone colour defines the left and right track boundaries. A neural-network detector such as YOLO finds cones and classifies them as blue, yellow, small orange or large orange; IIT Bombay reached 0.985 mAP on about 3,500 labelled cone images [2]. Depth then comes from the cone's size (mono), from keypoints and disparity (stereo), or—most accurately—by projecting LiDAR points into the camera bounding boxes. Cameras are cheap and dense but sensitive to lighting, rain and lens dirt, and their depth accuracy is poor at long range.

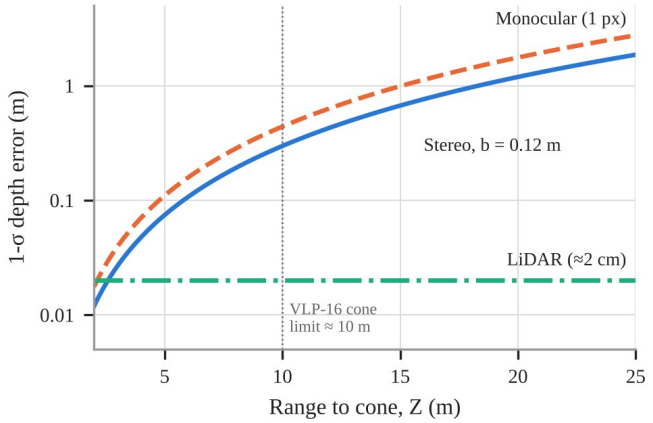


Fig. 2. Illustrative 1-σ depth error versus range for a small cone (0.325 m) with f = 700 px: monocular size-based ranging with 1 px height error, stereo with b = 0.12 m and 0.25 px disparity error (ZED 2i-like), and LiDAR range noise. The dotted line marks where a 2°-resolution 16-channel LiDAR can no longer guarantee a hit on the cone (Section VI).

D. Options and Cost

For prototyping, an Arducam OV9281 1 MP global-shutter module (MIPI CSI-2) costs ₹5,841 at Zbotic [12]; it is monochrome, so it suits keypoint and depth experiments but not colour classification. Competition cars use machine-vision cameras with global shutter and C-mount lenses (e.g. Basler ace or FLIR Blackfly S over USB3/GigE), bought through Indian machine-vision distributors on quotation. The most popular integrated stereo camera in FS is the Stereolabs ZED 2i (120 mm baseline, up to 2208 × 1242 per eye, 0.3–20 m depth, built-in IMU), at US$499 (≈₹47,400) from the manufacturer [3]. The Intel RealSense D435i is listed at ₹43,999.99 on ThinkRobotics (out of stock at the time of writing) [4].

VI. Light Detection and Ranging (LiDAR)

A. Principle of Operation

A LiDAR emits short infrared laser pulses (typically 905 nm, or 1550 nm for long-range units) and measures the time t until the reflection returns. The range is

$$R = c\,t\,/\,2 \tag{8}$$

so 1 cm of range corresponds to only 67 ps of round-trip time, which requires fast avalanche or single-photon detectors. Each return also carries an *intensity* (reflectivity) value. To build a 3-D *point cloud*, the beam is swept: classic **mechanical spinning** units (Velodyne, Ouster, Hesai) stack 16–128 lasers vertically and rotate them at 10–20 Hz to cover 360°. **Non-repetitive scanning** units such as the Livox Mid-360 steer the beam with rotating prisms, so coverage fills in over time, while **solid-state** units use MEMS mirrors or flash illumination with no macroscopic moving parts.

The decisive parameter for cone detection is *vertical angular resolution* Δθ. The vertical gap between neighbouring scan rings at range R is approximately R tan Δθ. For a 16-channel VLP-16 with Δθ = 2°, the gap at 10 m is 0.35 m—already larger than a 0.325 m cone—so beyond about 10 m a cone can fall completely between two rings. This matches the 10 m limit reported by IIT Bombay [2] and is why top teams use 32- to 128-channel sensors or mount the LiDAR low and tilted.

B. Use Case on an FS Driverless Car

LiDAR gives the most accurate cone positions (centimetre-level range noise, Fig. 2) independent of lighting. A typical pipeline removes the ground plane with RANSAC, clusters the remaining points (Euclidean clustering or DBSCAN), and rejects clusters whose size does not match a cone [2]. LiDAR alone cannot reliably tell blue from yellow, so it is fused with the camera: LiDAR-camera fusion gave IIT Bombay an average depth error of only 0.85 % [2]. Some teams also use the stripe pattern in the LiDAR intensity channel to infer cone colour. Drawbacks are cost, sparse returns at range, and the need for extrinsic calibration and time synchronisation with the cameras.

C. Options and Cost

Low-cost 2-D LiDARs such as the Slamtec RPLIDAR C1 (₹9,799.99) and YDLIDAR G2 (₹17,249.99) are sold by ThinkRobotics [16]; they scan a single plane at 8–12 m and are useful for learning ROS 2, SLAM and small-scale RC-car prototypes, but not for the full car. The Livox Mid-360 (360° × 59° field of view, 200,000 points/s, 40 m at 10 % reflectivity, ≤2 cm accuracy, 265 g) is an attractive budget 3-D option at about US$1,092 (≈₹1.04 lakh) from resellers [13]. The Velodyne VLP-16 was cut to about US$4,000 in 2018 [14] and the Ouster OS1-32 launched at about US$8,000 [15] (≈₹3.8 and ₹7.6 lakh respectively); 64- and 128-channel versions cost considerably more and are quote-based through distributors.

VII. Other Supporting Sensors

Several further sensors are needed to close the control loop. A **steering angle sensor** (absolute rotary encoder or potentiometer on the steering column or rack) is required for the steering actuator's position control and for the kinematic/bicycle model in the estimator. **Brake pressure transducers** confirm that the emergency brake system has actuated, which is safety-critical when no driver is on board. **Motor encoders/resolvers** on electric cars give wheel speed and torque information via the inverter. **Linear potentiometers** on the dampers and **temperature sensors** on tyres, battery and motors are used by vehicle-dynamics and powertrain teams. Automotive **radar** is robust to weather and measures velocity directly via Doppler, but its low angular resolution makes it a poor fit for small cones, so it is rarely used in FS.

VIII. Sensor Fusion and Recommended Suite

Table I summarises the strengths and weaknesses of each modality. The weaknesses are complementary: LiDAR is accurate but colour-blind, cameras see colour but range poorly, the IMU is fast but drifts, wheel speeds are absolute but fail under slip, and GNSS is absolute but slow and coarse. Fusion therefore happens at two levels: *perception fusion* (LiDAR points projected into camera detections) and *state fusion* (an

EKF or UKF combining IMU, wheel speed, GNSS and steering).

TABLE I
COMPARISON OF SENSOR MODALITIES

Sensor	Typical rate	Strength	Weakness
IMU	100–1000 Hz	Instant response to yaw and slip	Drift from bias integration
GNSS / RTK	1–20 Hz	Absolute position, accurate velocity	2 m (std.) error; outages
Wheel speed	Event-based	Cheap, absolute speed	Wrong under wheel slip
Mono camera	30–60 Hz	Colour, cheap, dense	Depth from size only
Stereo camera	15–100 Hz	Colour + depth	Error $\propto Z^2$; lighting
LiDAR	10–20 Hz	cm-level 3-D, lighting-independent	Cost; sparse at range; no colour

For all of this to work the sensors must share one clock and one geometry. Time synchronisation uses hardware triggering of the cameras and the IEEE 1588 Precision Time Protocol (PTP) or the GNSS pulse-per-second signal for the LiDAR and INS; extrinsic calibration (LiDAR-to-camera, IMU-to-vehicle) is done with targets before every event. All of the sensors discussed here have maintained ROS 2 drivers (e.g. the ZED, RealSense, Livox, Ouster, u-blox and SBG drivers), which matters because our stack is built on ROS 2. Table II lists representative products with indicative prices, and Table III proposes a two-tier suite: an affordable tier for learning and a small-scale test vehicle, and a competition tier for the full car.

TABLE II
REPRESENTATIVE SENSORS, INDICATIVE PRICES AND SOURCES

Sensor / model	Indicative cost	Source
IMU / INS		
MPU-6050 (GY-521), 6-axis	₹199	Probots [6]
BNO085 9-axis fusion IMU	₹2,279	Robocraze [7]
Xsens MTi-680G RTK GNSS/INS	≈₹3.4 lakh*	RS Components [8]
GNSS		
SparkFun GPS-RTK2 (u-blox ZED-F9P)	₹29,482	Robu.in [5]
Wheel speed		
LJ12A3-4-Z/BX inductive proximity	₹239	Probots [9]
Honeywell SNDH-T4L-G01 Hall-effect	₹4,500 + tax	IndiaMART [10]
Kistler Correvit SF-Motion (optical)	On quotation	Kistler [11]
Cameras		
Arducam OV9281 global shutter (mono)	₹5,841	Zbotic [12]
Intel RealSense D435i (active stereo)	₹43,999.99	ThinkRobotics [4]
Stereolabs ZED 2i (stereo)	≈₹47,400*	Stereolabs [3]
LiDAR		
Slamtec RPLIDAR C1 (2-D)	₹9,799.99	ThinkRobotics [16]
Livox Mid-360 (3-D)	≈₹1.04 lakh*	Reseller [13]
Velodyne VLP-16 (16-ch)	≈₹3.8 lakh*	2018 list [14]
Ouster OS1-32 (32-ch)	≈₹7.6 lakh*	Launch price [15]

*Converted from US$ or £ at ₹95/US$ and ₹127/£ (24 Oct 2026) [17], [18]; excludes shipping, customs duty and GST. Indian prices include GST unless stated. Prices and stock change frequently.

TABLE III
RECOMMENDED TWO-TIER SENSOR SUITE

Function	Learning / prototype tier	Competition tier
Inertial	BNO085 IMU	RTK GNSS/INS (Xsens MTi-680G, SBG Ellipse-N or VN-300)
Position	NEO-M8N GNSS	Dual-antenna RTK via the INS; ZED-F9P base
Wheel speed	Inductive proximity sensors on tone rings	Hall-effect sensors + inverter motor speed; optional optical ground-speed sensor
Vision	Global-shutter USB/CSI camera	ZED 2i stereo or two machine-vision cameras
Ranging	2-D LiDAR (RPLIDAR C1) or Livox Mid-360	32-ch or higher 3-D LiDAR (or Mid-360 mounted low)
Approx. total	≈₹0.2–1.3 lakh	≈₹5–12 lakh

IX. SOURCING IN INDIA

Sensors fall into three procurement channels. **Hobby and maker stores**—Robu.in (Pune), Robocraze and Probots (Bengaluru), ThinkRobotics and Zbotic—stock IMU breakouts, u-blox GNSS boards, proximity sensors, Raspberry-Pi-class cameras, 2-D LiDARs and occasionally RealSense cameras. They ship within a few days, accept online payment and provide GST invoices, which the team needs for reimbursement; warranties are short (typically 7–15 days, defects only [6], [9], [12]), so parts should be tested on arrival.

Component distributors and B2B marketplaces such as Mouser India, DigiKey, element14 India, RS Components and IndiaMART carry industrial parts like Honeywell Hall sensors and Xsens IMUs. Distributor prices for India usually include import handling, while IndiaMART sellers quote ex-tax and should be verified for authenticity and datasheet compliance.

Manufacturers and authorised distributors are the main route for competition-grade LiDARs, INS units, machine-vision cameras and optical speed sensors. These are quote-based; orders placed from abroad attract international shipping, customs duty and IGST on the landed value, and lead times of several weeks are common, so they must be ordered well before the testing season. Many manufacturers (Kistler, SBG, Ouster, Stereolabs, Xsens) run sponsorship or academic-discount programmes for Formula Student teams, which in practice is how many teams afford the competition tier.

X. CONCLUSION

A driverless FS car needs a complementary set of sensors rather than one perfect one. Cameras provide the cone colours that define the track; LiDAR provides accurate 3-D cone positions, limited by its vertical resolution; and the IMU, wheel speed sensors and GNSS together estimate the car's motion, with the IMU covering short time scales and the others correcting its drift. The simple calculations in this report—

LiDAR ring spacing, stereo error growth and IMU bias integration—explain the design choices seen in published FS systems. For a team starting out, a low-cost suite built from parts available on Indian online stores is enough to develop and test the full ROS 2 pipeline, after which the budget and sponsorship can be focused on a better 3-D LiDAR and a GNSS-aided INS, the two components that most directly limit performance on track.

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